• Critical Findings

- Critical: the intended rule layer is structurally incomplete. CSV mapping is optional at runtime, silently falls

  back to an empty set, and answerTemplate is loaded but never used, so the system relies mostly on broad keyword

  heuristics plus generic templates. See /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/

  configLoader.ts:57, /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/configLoader.ts:74, /C:/

  Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/keywordMatcher.ts:135.

- Critical: the fact model is far too small for the topics the engine claims to answer. CompanyData has no fields for

  legal entity/group structure, policy ownership, supplier due diligence, grievance mechanisms, wastewater discharge,

  waste routes, assurance scope, or compliance controls, yet those questions are still classified and answered. See /

  C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/types.ts:13, /C:/Users/User/Documents/CY/

  archive/Old/eu-tools-esg-rg/src/lib/respond/types.ts:31, /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/

  src/lib/respond/types.ts:50.

- Critical: multi-part questions are not decomposed or validated. The parser stores one free-text question, required

  is parsed but never enforced, and generation can answer a multi-part buyer question from the first 4-5 available

  points. See /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/questionParser.ts:207, /C:/

  Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/answerGenerator.ts:568, /C:/Users/User/

  Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/answerGenerator.ts:680.

- Critical: confidence is not evidence-based. Any matched data point can lift confidence; informal practices can

  upgrade unknown to estimated; users can manually flip confidence in the UI; readiness/export then count non-unknown

  items as answered. See /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/

answerGenerator.ts:699, /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/

answerGenerator.ts:754, /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/pages/ResultsPage.tsx:495, /C:/

Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/excelExporter.ts:295.

- Critical: multiple buyer-facing answers can sound authoritative without support:

  - legal/group structure from generic company basics via company/profile fallback: /C:/Users/User/Documents/CY/

    archive/Old/eu-tools-esg-rg/src/lib/respond/dataRetrieval.ts:155, /C:/Users/User/Documents/CY/archive/Old/eu-

    tools-esg-rg/src/lib/respond/answerGenerator.ts:260

  - anti-corruption/compliance from certifications or goals: /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-

    rg/src/lib/respond/dataRetrieval.ts:142, /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/

    respond/answerGenerator.ts:292, /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/

    answerGenerator.ts:327

  - Scope 3 methodology/boundary/category completeness from transport fragments: /C:/Users/User/Documents/CY/

    archive/Old/eu-tools-esg-rg/src/lib/respond/dataRetrieval.ts:88

  - wastewater/effluent controls from water withdrawal: /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/

    lib/respond/dataRetrieval.ts:49, /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/

    answerGenerator.ts:252

  - labor rights/human rights from workforce or diversity signals: /C:/Users/User/Documents/CY/archive/Old/eu-tools-

    esg-rg/src/lib/respond/keywordMatcher.ts:45, /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/

    respond/dataRetrieval.ts:120

  - generic policy/measure claims from informal-practice and matrix fallbacks: /C:/Users/User/Documents/CY/archive/

    Old/eu-tools-esg-rg/src/lib/respond/answerGenerator.ts:382, /C:/Users/User/Documents/CY/archive/Old/eu-tools-

    esg-rg/src/lib/respond/answerGenerator.ts:460

- LLM-polished answers with no post-validation against facts: /C:/Users/User/Documents/CY/archive/Old/eu-tools-

esg-rg/src/context/SessionContext.tsx:352, /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/context/

SessionContext.tsx:488

File-By-File Findings

- Critical: /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/answerGenerator.ts:460. Facts,

estimates, policy claims, maturity assumptions, and fallback narrative are mixed in one function. Templates make

unsupported assertions such as external audits, ongoing monitoring, proactive management systems, and formalization

roadmaps.

- Critical: /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/dataRetrieval.ts:88. Retrieval

maps very sparse fields into broad ESG domains, so one question can be answered with data meant for another.

transport is effectively a Scope 3 catch-all, and regulatory/goals collapse certifications plus sustainability goals

into governance claims.

- Critical: /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/keywordMatcher.ts:135. Matching is

keyword-score based, with no negative constraints, no sub-question model, and no requirement that matched data

exists for all requested components.

- High: /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/questionClassifier.ts:216. Ambiguous

questions default to MEASURE, which biases the engine toward plausible narrative instead of explicit insufficiency.

- High: /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/configLoader.ts:67. Canonical mapping

exists mostly as labels. answerTemplate is unused, and several metric key mappings are wrong or misleading:

diversion rate to recycled kilograms, TRIR to LTIFR, certifications to code of conduct. See /C:/Users/User/

Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/configLoader.ts:112.

- High: /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/questionParser.ts:112. Framework

  detection is applied wholesale across all parsed questions on a document; required is captured but never enforced

  downstream.

- High: /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/excelExporter.ts:129. Export makes the

  pack look submission-ready, counts estimated answers toward readiness, and presents methodology/confidence summaries

  that overstate actual evidence quality.

- High: /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/context/SessionContext.tsx:343. LLM enhancement

  can replace answers after generation without a second evidence/claim audit.

- Medium: /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/defensiveRewriter.ts:1. The rewriter

  removes cautionary language and converts explicit gaps into active roadmap prose, which is directionally unsafe for

  buyer-facing ESG responses.

- Medium: /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/components/ReadinessScore.tsx:63. Estimated

  answers are rewarded as partial completion, and “policy answered” means merely “not unknown,” not “supported by a

  policy artifact.”

Recommended Architectural Changes

- Split the model into four layers:
  - tracked facts: only user-entered or document-linked atomic facts with source, period, entity boundary, and

    completeness
  - calculated metrics: deterministic derivations with explicit formula, inputs, factor set, and uncertainty
  - policy/status claims: separate claim objects like has_policy, has_certification, scope3_methodology_defined,

    each requiring direct supporting facts

- fallback narrative: generated last, but only from approved facts/claims and with hard rules forbidding

  unsupported completion

- Replace “one question -> one narrative” with question -> required claim set/subparts -> evidence check -> answer

  assembly.

- Make confidence claim-level, not answer-level. An answer should be capped by the weakest required subpart, not

  lifted by any single strong datapoint.

- Add a hard “cannot answer safely” state distinct from unknown, and never rewrite that into optimistic roadmap copy

  unless the user is explicitly drafting internal improvement language rather than a buyer response.

- Prevent LLM rewrite unless it receives the approved fact/claim graph and must emit citations to those internal IDs.

Test Gaps

- Current tests mostly cover happy-path matching and basic energy/waste generation. They do not cover legal entity/

  group structure, supplier controls, human rights, wastewater, multi-part completeness, or governance overclaiming.

- I added /C:/Users/User/Documents/CY/archive/Old/eu-tools-esg-rg/src/lib/respond/__tests__/riskAudit.test.ts:1 with

  expected-failure tests for the highest-risk unsupported-answer paths.

- Additional tests still needed:

  - legal entity vs parent/subsidiary boundary separation

  - Scope 3 total vs methodology vs category coverage vs boundary

  - water withdrawal vs wastewater discharge/treatment/permit compliance

  - waste total vs hazardous waste vs disposal route vs trend

  - DEI percentages vs labor-rights due diligence/grievance/remedy

  - certification possession vs governance/compliance program claims

  - supplier onboarding vs ongoing due diligence vs modern slavery controls

- I could not execute the suite here because vitest is not installed in this checkout; npm test fails before running

  tests.

Suggested Refactor Plan

1. Introduce a canonical evidence-backed fact schema and migrate retrieval to emit facts only.
2. Define question requirement specs per topic/subpart, including disallowed substitutions.
3. Build deterministic calculators on top of facts and attach provenance.
4. Build a claim validator that decides which policy/status claims are actually supported.
5. Rewrite answer generation to consume only validated facts/claims and emit sectioned answers per required subpart.
6. Move confidence/readiness/export logic to the validated-claim layer, not raw draft text.
7. Keep the new audit tests as expected failures until each unsafe path is closed, then convert them to normal passing regressions.

ain patterns:

- Keyword misses and weighting errors are a major root cause: sites/facilities misses entirely; generic supplier/supply chain often beats the more specific buyer-requirements rules.
- Question classification is a second major root cause: several good templates are KPI/POLICY-only, but the question gets classified as MEASURE or KPI, so the wrong template wins.
- isDrafted is too aggressive for company/products/materials cases: answers with real stored profile fields still get flagged as drafted because the rule only checks operational/calculated points in the primary domain.
- Governance/reporting questions have several cross-template misfires: fines -> CSRD, reporting -> CSRD, assurance -> certifications, CSRD -> sustainability reporting.
- Training, human rights, and supply-chain monitoring are the weakest quality areas.

Legend: Y = materially data-backed, P = some real data used but incomplete/mis-scoped, N = no meaningful backing.

| Q# | Question | Domain | Template (line#) | Data-backed? | Issue | Suggested fix |

|---|---|---|---|---|---|---|

| 1 | Legal name/address/country | company / company_profile | Company profile (741) | P: legalEntityName,

registeredAddress, country | Responsive but over-answers and is drafted despite exact fields existing | Treat company

profile fields as data-backed; trim template to requested identity fields |

| 2 | Products/services/markets/customers | products / products_services | Products & services (808) | P:

productsServices, mainMarkets, customerTypes | Good answer but marked drafted because products data lives in company

points | Exempt company/products domains from auto-draft when explicit fields exist |

| 3 | Revenue band + ownership | financial_context / revenue | Revenue band (847) | P: revenueBand, ownership | Misses

ownership structure entirely | Add ownership to financial/company template |

| 4 | Sites + countries | null | None -> Unknown | N: numberOfSites, operatingCountries exist but never matched |

Plural keyword miss on sites/facilities | Add plural/stemmed site keywords or stemming |

| 5 | Parent/group structure | company / group_structure | Group structure (787) | P: parentCompany, subsidiaries,

ownership | Responsive but drafted despite exact fields | Same company-domain drafted fix |

| 6 | Electricity kWh | energy_electricity / energy_consumption | Electricity KPI (14) | Y: electricityKwh,

renewablePercent | None | None |

| 7 | Renewable electricity % | energy_electricity / renewable_share | Renewable % KPI (39) | Y: renewablePercent,

electricityKwh | Does not address “certified” source types | Add source-type fields or say certification/source split

not yet tracked |

| 8 | Natural gas + diesel | energy_fuel / energy_consumption | Fuel (1131) | Y: naturalGasM3, dieselLiters | Mostly

responsive | Optional: include gas kWh equivalent too |

| 9 | Scope 1 + methodology | emissions / scope_1 | GHG overview (95) | Y: auto-calc scope1/scope2 | Answers Scope 1

plus Scope 2; methodology/emission factors stay generic | Add Scope-1-specific template with factor-source text |

| 10 | Scope 2 loc/market | emissions / scope_2 | Scope 2 (126) | Y: scope2Location, scope2Market | Good | None |

| 11 | Scope 3 categories + total | emissions / scope_3 | Scope 3 (148) | N: no scope3/categories/travel data | Honest

but generic; no categories or total available | Add explicit “not yet quantified” KPI wording with any travel/commute

data if present |

| 12 | Energy efficiency measures + savings | energy_electricity / energy_efficiency | Efficiency (58) | P:

electricityKwh, renewablePercent; energySavingsKwh unused | Ignores existing energySavingsKwh=38,700; measures are

generic | Use energySavingsKwh; add actual measures/evidence fields |

| 13 | SBTi target + roadmap | goals / climate_targets | Goals summary (925) | P: sustainabilityGoal only | Answers

with policy name, not SBTi status, roadmap, or milestones | Let climate-target template (937) handle KPI-like target

questions too |

| 14 | Water withdrawal + sources | energy_water / water_usage | Water withdrawal (676) | Y: waterM3, municipal% |

Good | None |

| 15 | Water-stressed regions + measures | energy_water / water_stress | Water stress (720) | P: waterM3 only |

Generic inference; no site-level water-stress field | Add site water-risk field or explicitly say no formal WRI

screening data |

| 16 | Wastewater/treatment/permits | effluents / wastewater | Wastewater (698) | P: waterM3 only | Drafted generic

text; no treatment/process/permit data | Add wastewater/process/permit fields or return explicit gap |

| 17 | Total waste weight | waste / waste_total | Total waste (573) | Y: totalWasteKg, recyclingPercent,

hazardousWasteKg | Good | None |

| 18 | Recycling/diversion rate | waste / recycling | Recycling KPI (592) | Y: recyclingPercent, totalWasteKg | Good |

None |

| 19 | Hazardous waste + procedures | waste / hazardous_waste | Waste fallback (652) | P: hazardousWasteKg present but

template missed | Hazardous-waste KPI template blocked by MEASURE classification | Allow hazardous-waste template

(611) for MEASURE too |

| 20 | Packaging materials + recyclable/recycled % | packaging / packaging | Packaging (1311) | P: no packaging

fields; uses industry fallback + waste diversion | Answers material types, not requested percentages; drafted | Add

packaging % fields or explicitly say percentages not yet tracked |

| 21 | Circular economy initiatives | waste / circular_economy | Circular economy (634) | P: waste/diversion only |

Generic initiatives; no real take-back/design/material-recovery specifics | Add initiative-specific fields/evidence |

| 22 | Employee FTE | workforce / employee_count | Employee count (178) | Y: employeeCount | Good | None |

| 23 | Gender split | workforce / diversity | Gender KPI (197) | Y: femalePercent, employeeCount | Good | None |

| 24 | Women in leadership | workforce / leadership_diversity | Women leadership (401) | Y: womenInLeadershipPercent,

femalePercent | Good | None |

| 25 | Turnover rate | workforce / turnover | Turnover KPI (357) | Y: turnoverRate, employeeCount | Good | None |

| 26 | Collective bargaining % | workforce / collective_bargaining | Collective bargaining (377) | Y:

collectiveBargainingPercent, employeeCount | Good | None |

| 27 | Living wage + benchmarking | workforce / labor_practices | Fair wages (335) | Y: livingWageCompliant,

employeeCount | Good | None |

| 28 | DEI policy + initiatives/outcomes | workforce / dei_policy | DEI (241) | P: femalePercent,

womenInLeadershipPercent | Uses outcomes well, but there is no actual DEI policy field | Add explicit DEI-policy

field/document status |

| 29 | New hires + departures | workforce / hires_departures | Hires/departures (307) | P: newHires; departures

derived from turnover | Departures are estimated, not stored | Add actual departures field |

| 30 | TRIR + incidents + hours | health_safety / health_safety_kpi | H&S KPI (473) | Y: trir, lostTimeIncidents,

hoursWorked | Good | None |

| 31 | Lost-time incidents + LTIR | health_safety / health_safety_kpi | H&S KPI (473) | Y: lostTimeIncidents,

hoursWorked | Responsive, but still leads with TRIR | Split TRIR and LTIR templates or bias LTIR wording |

| 32 | Fatalities + corrective actions | health_safety / fatalities | Incident investigation (1103) | P: fatalities=0

exists but unused | Wrong template; omits explicit “0 fatalities” | Allow fatalities template (528) for MEASURE too |

| 33 | ISO 45001 + safety approach | regulatory / certifications | ISO 45001 (884) | P: ISO45001 cert only | Confirms

certification but omits safety management approach | Merge with H&S management template (547) |

| 34 | Incident investigation process | health_safety / incident_investigation | Incident investigation (1103) | Y:

TRIR/LTI + ISO45001 evidence | Good | None |

| 35 | Avg training hours + total + headcount | workforce / employee_count+training | Training summary (904) | Y:

trainingHoursPerEmployee, totalTrainingHours, totalFTE | Good | None |

| 36 | Types of training provided | health_safety / health_safety_management | H&S management (547) | P: trainingHours

exist, but wrong domain/template | Example health & safety drags question into H&S system answer | Add training-types

template; raise training rule for “types of training” |

| 37 | Career development + succession + effectiveness | training / training | Training summary (904) | P:

trainingHours only | Only gives hours/topics; misses development, succession, effectiveness | Add development/

effectiveness template |

| 38 | Human rights policy scope/communication | workforce / human_rights+freedom_of_association | Freedom of

association (266) | P: collectiveBargaining% only; human_rights draft policy ignored | Wrong question answered; misses

forced labor, child labor, scope, communication | Prioritize human-rights template (217) when those terms appear |

| 39 | Human-rights due diligence | workforce / human_rights | Sustainable procurement (1403) | N/P: no HR DD field;

generic supplier-procurement text | Answers different question | Add human-rights due-diligence measure template |

| 40 | Grievance mechanism + count | workforce / grievance | None -> structured fallback | N:

grievanceMechanismExists, grievancesReported exist but unused | KPI classification blocks grievance template; fallback

is useless | Let grievance template (444) support KPI or fix classifier |

| 41 | Freedom of association incidents | workforce / freedom_of_association | Freedom of association (266) | Y:

collectiveBargainingPercent, employeeCount | Good | None |

| 42 | Working conditions | workforce / working_conditions | Working conditions (287) | Y: employeeCount,

collectiveBargainingPercent, country | Good | None |

| 43 | Code of Conduct scope/comms/enforcement | goals / ethics | Ethics/code (960) | P: governance policy names only

| Misses communication and enforcement | Expand ethics template |

| 44 | Anti-corruption policy + training/monitoring/enforcement | goals / ethics | Ethics/code (960) | P: anti-

corruption policy exists | Same generic ethics answer; misses asked controls | Add anti-corruption-specific template |

| 45 | Fines/sanctions/legal proceedings | goals / fines_sanctions | CSRD (1068) | N/P: noSignificantFines exists but

unused | Wrong template via MEASURE classification + compliance overlap | Let fines template (1004) handle MEASURE or

tighten CSRD match |

| 46 | Data protection/GDPR | goals / data_protection | Data protection (1022) | Y: dataProtectionPolicy | Good | None

|

| 47 | Governance structure + ESG oversight | goals / governance | Governance oversight (979) | P: size, policies,

certs; no explicit oversight field | Reasonable but inferred from context rather than a stored governance field | Add

explicit ESG oversight/governance structure fields |

| 48 | ESG-linked executive comp | goals / esg_compensation | Exec comp (1047) | P: generic governance context only |

No actual compensation field; answer is generic but not drafted | Add explicit executive incentive field and draft

when absent |

| 49 | ESG risks + opportunities | swot / risk_management | ESG risk mgmt (1389) | N/P: generic only | Covers risks

more than opportunities; drafted | Add opportunity-specific content/fields |

| 50 | List all certifications + validity dates | regulatory / certifications | ISO 45001 (884) | P: all certs + valid

docs exist | Wrong template wins; only ISO 45001 returned | Prioritize general certs template (864) for “list all

certifications” |

| 51 | Publish ESG report + framework | regulatory / csrd+transparency | CSRD (1068) | P: publishesReport=false,

reportingFramework empty, csrdApplicable=assessing | Wrong template; should answer report publication first |

Prioritize reporting template (1374) for publish/report wording |

| 52 | External assurance/verification | regulatory / certifications | General certs (864) | N/P:

externalAssurance=false, assuranceStandard empty | Answers certifications instead of assurance status | Add assurance

template using externalAssurance / assuranceStandard |

| 53 | CSRD applicability + timeline | regulatory / csrd | Sustainability reporting (1374) | P:

csrdApplicable=assessing exists | Wrong template because CSRD template excludes KPI | Let CSRD template (1068) support

KPI too |

| 54 | Supplier Code of Conduct + standards | materials / supplier_management | Supplier code (1256) | P:

supplier_code policy exists, but answer says being developed | Generic supply chain rule beats supplier-code rule;

drafted incorrectly | Raise supplier-code rule weight and/or stem supplier keywords |

| 55 | % suppliers ESG-assessed | buyer_requirements / supplier_code | Supplier assessment (1333) | Y:

suppliersAssessedPercent | Good | None |

| 56 | Ongoing supply-chain ESG monitoring | materials / supplier_management | Sustainable procurement (1403) | P:

suppliersAssessedPercent exists but unused | Answers procurement criteria, not monitoring tools/screening | Bias to

buyer-requirements monitoring template (1277) |

| 57 | Responsible sourcing + CMRT/EMRT | materials / raw_materials+conflict_minerals | Raw materials (1241) | P: no

CMRT field; supplier context only | Omits conflict-minerals due diligence and sourcing policy status | Let conflict-

minerals template (1292) or new combined template win |

| 58 | Supplier ESG non-compliance escalation | materials / supplier_management | Sustainable procurement (1403) | N/

P: no escalation field; wrong template | Supplier non-compliance template exists but never fires | Raise buyer-

requirements for non-compliance/escalation keywords |

| 59 | Scope 3 transport emissions + fleet/travel/commute | transport / scope_3+fleet+transport | Transport/logistics

(1194) | N: no transport/scope3 data | Generic logistics narrative instead of explicit data gap | Add KPI transport

fallback that states not tracked and lists missing fields |

| 60 | Measures to reduce transport emissions | transport / fleet+transport | Transport/logistics (1194) | N/P: no

measure fields | Generic logistics statement; no current measures cited | Add transport-reduction measure template |

Highest-priority fixes from an output-quality standpoint are Q4, Q12, Q19, Q32, Q36, Q38-Q40, Q45, Q50-Q54, and Q56-

Q60.

ESG Response Generator — Structural Audit

CRITICAL FINDINGS

C1. Scrub rules silently convert admissions of missing data into false forward-looking claims

File: response-ready/domain-packs/esg/scrubRules.ts:19-26

Severity: CRITICAL

The defensive rewriter transforms honest gap disclosures into fabricated commitments:

Input text	Output text
"we do not track water consumption"	"we are establishing tracking for water consumption"
"we lack a formal policy"	"we are developing a formal policy"
"there is no grievance mechanism"	"we are establishing grievance mechanism"
"data was not collected"	"data collection is currently being established"

Problem: If a template or matrix generator honestly says the company doesn't track something, the scrub rules rewrite it to claim the company is actively building the capability. A buyer reading "we are establishing tracking for Scope 3 emissions" reasonably concludes work has begun — but the system has no evidence of that. This is the single highest-risk pattern in the entire system: it manufactures forward-looking commitments from nothing.

Fix: Remove rules 13-18 entirely. Replace with a structured dataGaps section in the answer that separates "what we have" from "what we don't have" without claiming work-in-progress status.

C2. Matrix generator produces plausible policy/measure claims without any underlying data

File: response-ready/domain-packs/esg/matrixGenerator.ts:69-74

Severity: CRITICAL

When maturityBand === 'none' (no data, no informal practices), the POLICY matrix generates:

"[Company] is committed to responsible management in this area."

"We are developing a formalised policy for publication in [nextYear]."

"In the interim, our approach is guided by [industry management approach]."

This answer asserts:

1. The company has a commitment (no evidence)

2. A policy is being developed with a date (fabricated timeline)

3. The company follows industry practices (no evidence)

The guard in answerGenerator.ts:265 (shouldUseMatrix = hasOperationalData || maturityBand === 'informal' ||

maturityBand === 'formal') is supposed to block this, but the maturityResolver at matrixGenerator.ts:23-36 returns

'formal' whenever hasData === true. The hasData check in answerGenerator.ts:230 considers ANY data point (including

company name, industry, country) as "having data." So a company with only a name and country filled in gets

hasData=true → maturityBand='formal' → matrix generates confident formal-tier answers about policies that don't exist.

Fix: hasData for maturity purposes must check only operational/calculated points in the relevant domain, not generic

company context.

C3. Confidence scoring counts irrelevant data points as evidence

File: response-ready/src/engine/answerGenerator.ts:143-158

Severity: CRITICAL

determineConfidence() counts ALL points — including company name, industry, country — when deciding confidence. A

question about Scope 3 emissions can get confidence: 'medium' if the only data available is the company's name and

country (which are injected into every domain's company points by dataModel.ts).

The check is:

```
const allPoints = [...context.company, ...context.operational, ...context.calculated];
if (allPoints.length > 0) return 'low'; // ← This fires for every question
```

Every question gets at least 'low' confidence because company profile data is always injected. This means the system

will never show confidence: 'none' for any question as long as a company name exists.

Fix: determineConfidence should only count operational/calculated points that belong to the question's primary domain.

C4. Template answers can use data from a different ESG domain

File: response-ready/src/engine/answerGenerator.ts:298-301

Severity: HIGH

Phase 4 (structured data fallback) filters points by primary domain OR secondaryDomains.slice(0, 1) — but the dataMap

passed to templates in Phase 1 contains ALL points from ALL categories (company + operational + calculated),

regardless of domain. A template for "waste management" can read and display energy data if the field names overlap or

if the template author uses num(dm, 'totalElectricity').

The buildDataMap function at line 58-64 flattens everything into a single Map keyed by field name, discarding domain

information.

Fix: Either scope the dataMap to the matched domain, or add domain guards to individual templates.

C5. Zero values treated as "no data" throughout the bridge

File: esg-passport/src/lib/dataBridge.js:144-176

Severity: HIGH

The dataBridge uses || undefined for numeric fields:

electricityKwh: totals.electricityKwh || undefined,

totalWasteKg: totals.totalWasteKg || undefined,

fatalities: totals.fatalities || undefined,

If a company has zero fatalities, 0 || undefined returns undefined, which means:

- The fatality template won't fire (no data)
- The answer becomes "Unknown — input required" instead of "No fatalities during the period"
- Same for zero waste, zero incidents, zero hazardous waste

This is especially dangerous for H&S questions where zero is the desired answer.

Fix: Use != null checks instead of truthiness: totals.fatalities != null ? totals.fatalities : undefined

---

FILE-BY-FILE FINDINGS

response-ready/domain-packs/esg/dataModel.ts

#	Finding	Severity

D1	ownership field is declared twice (lines 28, 31) — TypeScript silently takes last	LOW
	esgRetrieveData injects company context (name, industry, country, FTE) into every domain	
D2	including unrelated ones like effluents and packaging. This inflates point counts and distorts	HIGH
	confidence.	
D3	Auto-calculated Scope 1/2 emissions are injected with confidence: 'medium' but the label says	MEDIUM
	"Auto-calculated" — templates don't consistently distinguish user-provided vs auto-calculated.	
	POLICY_CATEGORY_DOMAINS maps environmental policies to energy_electricity, energy_fuel,	
D4	energy_water, emissions, waste, effluents — but a generic "Environmental Policy" gets injected	HIGH
	into all 6 domains, making it seem like the company has specific policies for each.	

response-ready/domain-packs/esg/answerTemplates.ts

#	Finding	Severity

		Energy efficiency MEASURE template (line 59-73) generates an answer even when has(dm,	
T1	'totalElectricity') is false — it hardcodes "Key efficiency measures include regular energy	HIGH	
	audits, equipment maintenance schedules, LED lighting upgrades" with no evidence.		
		H&S management template (line ~22-23 area) hardcodes "hazard identification, risk assessment,	
T2	continuous improvement, inspections, training, emergency preparedness" regardless of whether	HIGH	
	any of this is documented.		
T3	Corporate governance template assumes quarterly reporting cadence without evidence.	MEDIUM	
T4	Freedom of association template states "no incidents of restriction" without checking — assumes	HIGH	
	compliance by default.		
T5	Ethics/code of conduct template for the "developing" case says "formalizing anti-corruption and	MEDIUM	
	fair business practices" — claims specific practice areas with no evidence.		
T6	Working conditions MEASURE template describes "structured approach to working hours, overtime,	HIGH	

| | rest periods, leave entitlements" — asserts operational controls that may not exist. | |

| T7 | Wastewater template for non-heavy industries states "all wastewater discharged to municipal | MEDIUM |

| | treatment" — factual claim about infrastructure with no verification. | |

response-ready/domain-packs/esg/matrixGenerator.ts

| # | Finding | Severity |

| M1 | All maturity bands produce affirmative text. Even 'none' generates "committed to responsible | CRITICAL |

| | management" + "developing formalised policy" — there is no honest "we don't have this" output. | |

| M2 | KPI 'none' band says "Setting up data collection for [nextYear] disclosure cycle" — fabricates | HIGH |

| | a timeline. | |

| M3 | MEASURE 'none' band says "Planned initiatives for [nextYear]" and lists industry library | HIGH |

| | measures — presents generic industry practices as company plans. | |

response-ready/domain-packs/esg/informalPractices.ts

#	Finding	Severity
I1	generateAnswer receives empty strings for companyName and industry (answerGenerator.ts:287-288) when called from the engine. The handler falls back to "Our organization" but loses industry context.	MEDIUM
I2	Appends "management approach encompasses [industry approach]" — presents generic industry knowledge as company behavior.	HIGH

response-ready/src/engine/answerGenerator.ts

#	Finding	Severity

| A1 | Phase 1 template match can select a secondary-domain template when primary domain has a better | LOW |
| | match lower in the candidate list, due to sort stability. | |

| A2 | drafted flag logic: `drafted = explicitDrafted | |

	confidenceSource for drafted answers is 'drafted' but answerConfidence is forced to 'medium'	
A3	(line 338). A completely fabricated template answer with no data gets "medium confidence,	HIGH
	drafted" — the UI shows this as amber/acceptable rather than red/warning.	

response-ready/src/engine/keywordMatcher.ts

#	Finding	Severity
K1	CSV rules tried first with no priority comparison against keyword rules — a CSV rule can override a better keyword match.	LOW
K2	Tie-breaking between equal-scoring matches is non-deterministic (depends on array order). Two	LOW

| | questions about slightly different Scope 3 topics could randomly swap templates. | |

response-ready/src/engine/questionClassifier.ts

#	Finding	Severity
	Category text is prepended to question text for scoring. A question like "POLICY: What are your	
Q1	Scope 1 emissions?" would score high on POLICY just from the category prefix, even though the	MEDIUM
	actual question is a KPI request.	

esg-passport/src/lib/dataBridge.js

#	Finding	Severity
B1	parseInt(profile?.totalEmployees) with no radix and no NaN guard — if profile has "50+" this	LOW
	silently becomes 50.	

B2	Reporting year auto-detect picks the most recent year with ANY data record. If a user enters one test record in 2026, it ignores the complete 2025 dataset.	MEDIUM	
B3	`grievancesReported: totals.grievancesReported		
B4	scope3Tco2e: totals.scope3Total — field name mismatch between bridge (scope3Total) and engine (scope3Tco2e). If getAnnualTotals doesn't have a scope3Total field, Scope 3 is silently lost.	MEDIUM	

esg-passport/src/lib/store.js

#	Finding	Severity
S1	getAnnualTotals uses sum() for scope1Tco2e, scope2Tco2e — adds monthly emissions values. If user enters annual totals in one month, every other month contributes 0 (correct). But if they enter partial monthly values, the sum may double-count.	MEDIUM

| S2 | avg() for renewablePercent averages across months. If data exists for only 2 of 12 months, the | MEDIUM |

| | average includes only those 2 months — but the result looks like an annual figure. | |

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RECOMMENDED ARCHITECTURAL CHANGES

1. Separate tracked facts from generated claims

Currently the system blurs the line between:

- Tracked facts (user entered 50,000 kWh)
- Calculated metrics (Scope 2 = 20.85 tCO2e from kWh × grid factor)
- Policy/status claims ("We maintain an environmental policy")
- Fabricated narrative ("We are developing a formalised policy for 2025")

Proposed separation:

```
DataContext {
  tracked: RetrievedDataPoint[]      // User-entered, verified
  calculated: RetrievedDataPoint[]   // Derived with methodology
  attested: RetrievedDataPoint[]     // Policy/cert claims with document evidence
  narrative: string[]                // Generated prose — clearly labeled as draft
  gaps: GapDeclaration[]             // What's missing, without false promises
}
```

Each answer should carry provenance tags showing which parts come from which tier.

2. Kill the "passive → active roadmap" scrub rules

Replace rules 13-18 in scrubRules.ts with a structured gap disclosure format:

"[DATA GAP] We do not currently track [metric].

Collection target: [user-set date or 'Not yet planned']."

The user should set their own roadmap dates in the Passport; the system should never invent them.

3. Domain-scoped confidence scoring

determineConfidence should:

1. Only count points where point.domain === matchResult.primaryDomain

2. Exclude company profile points (name, industry, country) from evidence counts

3. Return 'none' when only company context exists

4. Fix zero-value suppression in dataBridge

Replace all value || undefined patterns with value != null ? value : undefined for numeric fields. Critical for:

fatalities, grievancesReported, hazardousWasteKg, lostTimeIncidents, totalWasteKg, recyclingPercent.

5. Add a "claim registry" to templates

Each template should declare what facts it asserts:

```
{
  claims: [
    { type: 'metric', field: 'totalElectricity', required: true },
    { type: 'practice', description: 'energy audits', evidence: 'none' },
  ]
}
```

At generation time, any claim without backing data should be flagged as unsupported_claim rather than emitted as

confident prose.

TEST GAPS

High-risk topics with no dedicated tests:

Topic	Risk	Current Coverage
Scope 3 emissions	Fabricated category lists, false measurement claims	None
Legal entity / group structure	Parent/subsidiary claims from free text	None
Fatalities / H&S KPIs	Zero-value suppression (C5), false "no fatalities" vs "no data"	None
Labor rights (forced labor, child labor)	Policy template asserts compliance without evidence (T4)	None
Grievance mechanism	Zero-grievance suppression (B3)	None

Fines / sanctions	noSignificantFines flag trusted without verification	None
Certifications	Expired certs could be presented as current	None
CSRD applicability	Legal determination presented as self-assessment	None
Scrub rule transforms	No test that rule transforms are safe/accurate	None
Matrix generator outputs	No test that maturity=none doesn't fabricate claims	None
Cross-domain data leakage	No test that energy data can't appear in waste answers	None
dataBridge zero handling	dataBridge.test.js doesn't test zero-value fields	Partial

Recommended new test files:

1. scrubRules.test.ts — Verify each rule doesn't turn honest gaps into false claims

2. matrixGenerator.test.ts — Verify maturity='none' outputs carry drafted flag and appropriate confidence

3. answerTemplates.test.ts — For each template: verify it returns null when required fields are missing; verify it

doesn't assert unsubstantiated practices

4. dataBridge-zeros.test.js — Verify fatalities: 0, grievancesReported: 0, hazardousWasteKg: 0 survive the bridge

5. confidenceScoring.test.ts — Verify company-profile-only contexts get 'none' confidence

6. crossDomainLeakage.test.ts — Verify a waste question can't read/display energy data

---

SUGGESTED REFACTOR PLAN

Phase 1 — Stop the bleeding (safety fixes, no architecture change)

1. Fix zero-value suppression in dataBridge.js (C5, B3)

2. Remove or neuter scrub rules 13-18 (C1)

3. Scope determineConfidence to primary domain points only (C3)

4. Fix hasData check to exclude company profile context (C2)

5. Add drafted flag to all matrix generator 'none' band outputs

Phase 2 — Add guardrails (new tests for existing behavior)

1. Write scrub rule safety tests

2. Write template assertion tests (what does each template claim without evidence?)

3. Write dataBridge zero-value tests

4. Write cross-domain leakage tests

5. Write matrix generator fabrication tests

Phase 3 — Structural separation (architecture refactor)

1. Split DataContext into tracked/calculated/attested/narrative tiers

2. Add claim registry to templates

3. Make templates return structured { text, claims[], evidence[] } instead of plain strings

4. Add a post-generation audit step that flags unsupported claims

5. Replace "active roadmap" scrub rules with structured gap declarations

Phase 4 — Confidence overhaul

1. Per-claim confidence (not per-answer)

2. Propagate data point confidence through calculations (calculated = min of inputs)

3. Add evidenceType to each answer section: 'measured' | 'calculated' | 'attested' | 'assumed' | 'fabricated'

4. Export shows per-section evidence provenance

Here's the complete audit of all 60 questions in the comprehensive_buyer template.

Hartmann Demo — Computed Annual Data (2025)

First, the engine-ready values after dataBridge transformation:

Field	Value	Source
companyName	Hartmann Precision	tradingName
industry	Manufacturing	profile
country	Germany	DE → lookup
employeeCount	283	last month

numberOfSites	2	profile
electricityKwh	2,321,000	sum
renewablePercent	52	avg
naturalGasM3	95,640	1,009,000 kWh ÷ 10.55
dieselLiters	33,950	sum (vehicleFuelLiters)
energySavingsKwh	38,700	sum
waterM3	8,080	sum
waterSourceMunicipalPercent	95	avg
totalWasteKg	157,500	sum

recyclingPercent	80	recycledKg/totalKg
hazardousWasteKg	8,370	sum
femalePercent	27	77/283
womenInLeadershipPercent	29	avg
turnoverRate	10.9	avg
collectiveBargainingPercent	69	avg
livingWageCompliant	true	profile
grievanceMechanismExists	true	profile
grievancesReported	4	sum

newHires	27	sum
suppliersAssessedPercent	42	avg
trainingHoursPerEmployee	16.3	4,620/283
trirRate	1.49	(4/536,800)×200,000
lostTimeIncidents	1	sum
fatalities	0	sum
hoursWorked	536,800	sum
scope1Tco2e	auto-calc	from fuel data
scope2Tco2e	auto-calc	from electricity+country
certifications	ISO 9001, ISO 14001, ISO 45001	profile+policies

noSignificantFines	none	profile
dataProtectionPolicy	true	profile
publishesSustainabilityReport	false	profile
csrdApplicable	assessing	profile
Active policies	environmental_policy, environmental, climate_ghg, health_safety, code_of_conduct, anti_corruption, supplier_code, energy_management	
Draft policies	human_rights, data_protection	
Documents	ISO 14001/45001/9001 certs, Energy Audit, Haz Waste Manifest	

Full 60-Question Audit

Legend

- Data-backed: Y = uses real operational data, P = uses company profile/policy only, N = generic narrative (drafted)
- Issue severity: -- = none, LOW = minor, MED = notable gap, HIGH = answers wrong question or misses key data

Q#: 1

Question (abbreviated): Legal name, address, country

Domain: company

Topics: company_profile

Template (line): L740 company profile

Data?: Y

Issue: --

Fix:

Q#: 2

Question (abbreviated): Products, markets, customers

Domain: products

Topics: products_services

Template (line): L808 products

Data?: Y

Issue: --

Fix:

Q#: 3

Question (abbreviated): Revenue band + ownership

Domain: financial_context

Topics: revenue

Template (line): L847 revenue

Data?: Y

Issue: MED: Only answers revenue, ignores ownership question

Fix: Revenue template should pull ownership from company data, or keyword match should include company as secondary

domain

Q#: 4

Question (abbreviated): Sites/facilities, countries

Domain: site

Topics: facilities

Template (line): L763 sites

Data?: Y

Issue: --

Fix:

Q#: 5

Question (abbreviated): Subsidiary / parent company

Domain: company

Topics: group_structure

Template (line): L788 group structure

Data?: Y

Issue: --

Fix:

Q#: 6

Question (abbreviated): Total electricity (kWh)

Domain: energy_electricity

Topics: energy_consumption

Template (line): L15 electricity KPI

Data?: Y

Issue: --

Fix:

Q#: 7

Question (abbreviated): % renewable electricity

Domain: energy_electricity

Topics: renewable_share

Template (line): L39 renewable % KPI

Data?: Y

Issue: --

Fix:

Q#: 8

Question (abbreviated): Natural gas + diesel

Domain: energy_fuel

Topics: energy_consumption

Template (line): L1130 fuel

Data?: Y

Issue: LOW: Gas in m^3 only, question accepts kWh too

Fix: Template could show both units

Q#: 9

Question (abbreviated): Scope 1 + methodology

Domain: emissions

Topics: ghg_emissions, scope_1

Template (line): L96 GHG overview

Data?: Y(calc)

Issue: LOW: Says "standard emission factors" but doesn't name which ones

Fix: Template should name the factors (DEFRA, IEA, etc.)

Q#: 10

Question (abbreviated): Scope 2 location + market

Domain: emissions

Topics: scope_2, ghg_emissions

Template (line): L127 scope 2

Data?: Y(calc)

Issue: --

Fix:

Q#: 11

Question (abbreviated): Scope 3 — do you measure?

Domain: emissions

Topics: scope_3, ghg_emissions

Template (line): L148 scope 3

Data?: N

Issue: -- Correctly says "not yet quantified"

Fix:

Q#: 12

Question (abbreviated): Energy efficiency measures + savings

Domain: energy_electricity

Topics: energy_efficiency

Template (line): L58 efficiency MEASURE

Data?: P

Issue: MED: Demo has energySavingsKwh=38,700 but template ignores it

Fix: Template should use energySavingsKwh field: "saving X kWh this period"

Q#: 13

Question (abbreviated): SBTi targets / decarb roadmap

Domain: goals

Topics: climate_targets

Template (line): L937 climate targets POLICY

Data?: P

Issue: LOW: Doesn't address "roadmap" or "interim milestones"

Fix: Could add roadmap framing when emissions baseline exists

---

Q#: 14

Question (abbreviated): Water withdrawal + sources

Domain: energy_water

Topics: water_usage

Template (line): L677 water KPI

Data?: Y

Issue: --

Fix:

Q#: 15

Question (abbreviated): Water-stressed regions?

Domain: energy_water

Topics: water_stress

Template (line): L720 water stress

Data?: P

Issue: -- Appropriately says "not in high stress regions"

Fix:

Q#: 16

Question (abbreviated): Wastewater management

Domain: effluents

Topics: wastewater

Template (line): L698 wastewater

Data?: P

Issue: --

Fix:

Q#: 17

Question (abbreviated): Total waste (kg/tonnes)

Domain: waste

Topics: waste_total

Template (line): L573 waste KPI

Data?: Y

Issue: --

Fix:

Q#: 18

Question (abbreviated): Recycling/diversion rate

Domain: waste

Topics: recycling

Template (line): L592 recycling KPI

Data?: Y

Issue: --

Fix:

Q#: 19

Question (abbreviated): Hazardous waste + management

Domain: waste

Topics: hazardous_waste

Template (line): L611 hazardous KPI

Data?: Y

Issue: --

Fix:

Q#: 20

Question (abbreviated): Packaging materials + % recyclable

Domain: packaging

Topics: packaging

Template (line): L1311 packaging

Data?: N

Issue: LOW: No packaging data in model; narrative is industry-aware but no numbers

Fix: Would need packaging fields in data model

Q#: 21

Question (abbreviated): Circular economy initiatives

Domain: waste

Topics: circular_economy

Template (line): L634 circular MEASURE

Data?: P

Issue: --

Fix:

Q#: 22

Question (abbreviated): Employee FTE count

Domain: workforce

Topics: employee_count

Template (line): L178 employee KPI

Data?: Y

Issue: --

Fix:

Q#: 23

Question (abbreviated): Gender breakdown (% F/M)

Domain: workforce

Topics: diversity

Template (line): L198 gender KPI

Data?: Y

Issue: --

Fix:

Q#: 24

Question (abbreviated): % women in leadership

Domain: workforce

Topics: diversity, leadership_diversity

Template (line): L401 leadership KPI

Data?: Y

Issue: --

Fix:

Q#: 25

Question (abbreviated): Employee turnover rate

Domain: workforce

Topics: turnover

Template (line): L357 turnover KPI

Data?: Y

Issue: --

Fix:

Q#: 26

Question (abbreviated): Collective bargaining %

Domain: workforce

Topics: collective_bargaining

Template (line): L377 CBA KPI

Data?: Y

Issue: --

Fix:

Q#: 27

Question (abbreviated): Living wage + benchmarking

Domain: workforce

Topics: labor_practices

Template (line): L336 fair wages MEASURE

Data?: Y

Issue: --

Fix:

Q#: 28

Question (abbreviated): DEI policy + initiatives + outcomes

Domain: workforce

Topics: dei_policy

Template (line): L241 DEI POLICY

Data?: P

Issue: LOW: Gives metrics but not "specific initiatives" or "measurable outcomes" as asked

Fix: Template could reference industry-specific DEI actions

Q#: 29

Question (abbreviated): New hires + departures

Domain: workforce

Topics: hires_departures

Template (line): L307 hires/departures KPI

Data?: Y

Issue: --

Fix:

Q#: 30

Question (abbreviated): TRIR + recordable incidents + hours worked

Domain: health_safety

Topics: health_safety_kpi

Template (line): L473 H&S KPI

Data?: Y

Issue: MED: Gives TRIR and hours worked but not the raw recordable incident count (4)

Fix: dataBridge should pass recordableIncidents; template should show it

---

Q#: 31

Question (abbreviated): Lost time incidents + LTIR

Domain: health_safety

Topics: health_safety_kpi

Template (line): L473 H&S KPI

Data?: Y

Issue: --

Fix:

Q#: 32

Question (abbreviated): Work-related fatalities

Domain: health_safety

Topics: fatalities

Template (line): L528 fatalities KPI

Data?: Y

Issue: --

Fix:

Q#: 33

Question (abbreviated): ISO 45001 + safety management approach

Domain: regulatory

Topics: certifications, h&s_mgmt

Template (line): L884 ISO 45001

Data?: Y

Issue: MED: Confirms cert but doesn't "describe your safety management approach"

Fix: Merge L884 (cert check) + L547 (management system description) into one template, or fall through to L547 when

cert is confirmed

Q#: 34

Question (abbreviated): Incident investigation process

Domain: health_safety

Topics: incident_investigation

Template (line): L1103 investigation MEASURE

Data?: Y

Issue: --

Fix:

Q#: 35

Question (abbreviated): Training hours/employee + totals

Domain: training

Topics: training

Template (line): L903 training

Data?: Y

Issue: --

Fix:

Q#: 36

Question (abbreviated): Types of training provided

Domain: training

Topics: training

Template (line): L903 training

Data?: P

Issue: MED: Same template as Q35; only says "H&S, technical, sustainability" generically

Fix: Need a MEASURE-typed training template describing programme categories in more detail

---

Q#: 37

Question (abbreviated): Career development + succession planning + training effectiveness

Domain: training

Topics: training

Template (line): L903 training

Data?: P

Issue: HIGH: Template answers "training hours delivered" — completely wrong question. Career development, succession

planning, effectiveness measurement not addressed at all

Fix: New template for training/development domain with topics like career_development, training_effectiveness

---

Q#: 38

Question (abbreviated): Human rights policy (forced labor, child labor, freedom of association)

Domain: workforce

Topics: human_rights

Template (line): L217 HR POLICY

Data?: P

Issue: MED: Cites H&S policy as basis for human rights commitments (misleading — actual human_rights policy is in

draft)

Fix: Policy injection should distinguish social policy categories; or template should check for actual HR-named

policies

Q#: 39

Question (abbreviated): Human rights due diligence processes

Domain: workforce

Topics: human_rights

Template (line): none (L217 is POLICY-only)

Data?: N

Issue: HIGH: No MEASURE template for human rights due diligence. Falls to Phase 4 data dump or "no data"

Fix: Add a MEASURE-typed human rights due diligence template

Q#: 40

Question (abbreviated): Grievance mechanism + count

Domain: workforce

Topics: grievance

Template (line): L444 grievance POLICY

Data?: Y

Issue: --

Fix:

Q#: 41

Question (abbreviated): Freedom of association + incidents

Domain: workforce

Topics: freedom_of_association

Template (line): L267 freedom POLICY/MEASURE

Data?: Y

Issue: --

Fix:

Q#: 42

Question (abbreviated): Working conditions (hours, overtime, rest)

Domain: workforce

Topics: working_conditions

Template (line): L287 working conditions

Data?: P

Issue: --

Fix:

Q#: 43

Question (abbreviated): Code of Conduct + enforcement

Domain: goals

Topics: ethics

Template (line): L960 ethics POLICY

Data?: Y

Issue: LOW: Doesn't address "how compliance is enforced"

Fix: Could add enforcement framing

Q#: 44

Question (abbreviated): Anti-corruption — training, monitoring, enforcement

Domain: goals

Topics: ethics

Template (line): L960 ethics POLICY

Data?: Y

Issue: MED: Same template as Q43; doesn't address anti-corruption training, monitoring, or enforcement mechanisms

Fix: Need dedicated anti-corruption template or extend ethics template to cover these when "anti-corruption" keywords

present

Q#: 45

Question (abbreviated): Fines / sanctions / legal proceedings

Domain: goals

Topics: fines_sanctions

Template (line): L1004 fines POLICY

Data?: Y

Issue: --

Fix:

Q#: 46

Question (abbreviated): Data protection / GDPR

Domain: goals

Topics: data_protection

Template (line): L1022 data protection

Data?: Y

Issue: --

Fix:

Q#: 47

Question (abbreviated): Corporate governance / ESG oversight

Domain: goals

Topics: governance, strategy

Template (line): L979 governance POLICY

Data?: Y

Issue: --

Fix:

Q#: 48

Question (abbreviated): ESG-linked executive compensation

Domain: goals

Topics: esg_compensation

Template (line): L1047 ESG comp POLICY

Data?: P

Issue: -- Honestly says "no formal structure yet"

Fix:

Q#: 49

Question (abbreviated): Material ESG risk management

Domain: swot

Topics: risk_management

Template (line): L1389 ESG risk

Data?: N

Issue: LOW: Generic narrative, no company-specific risks

Fix: Could reference known cert-backed areas as managed risks

---

Q#: 50

Question (abbreviated): All certifications + validity dates

Domain: regulatory

Topics: certifications

Template (line): L865 certifications

Data?: Y

Issue: MED: (1) Duplicate cert names from profile+documents; (2) Validity dates exist in document records but template

says "available on request"

Fix: Dedupe cert names properly; pull validUntil from documents and display them

---

Q#: 51

Question (abbreviated): Sustainability report + framework

Domain: regulatory

Topics: transparency

Template (line): L1375 reporting

Data?: P

Issue: LOW: Template always says "does not currently publish" without checking publishesSustainabilityReport flag.

Correct for Hartmann but latent bug.

Fix: Template should check the flag and reportingFramework field

---

Q#: 52

Question (abbreviated): External verification / assurance

Domain: regulatory?

Topics: weak match

Template (line): likely certifications fallback

Data?: N

Issue: HIGH: No keyword rule for "externally verified/assured" and no template for external assurance.

externalAssurance=false and assuranceStandard data exist but are unused.

Fix: Add keyword rule for verification/assurance → regulatory domain with 'verification' topic; add dedicated template

---

Q#: 53

Question (abbreviated): CSRD obligations + timeline

Domain: regulatory

Topics: csrd

Template (line): L1068 CSRD

Data?: Y

Issue: --

Fix:

Q#: 54

Question (abbreviated): Supplier Code of Conduct

Domain: buyer_requirements

Topics: supplier_code

Template (line): L1256 supplier code POLICY

Data?: N

Issue: HIGH: supplier_code policy IS active but buyer_requirements domain is not in POLICY_CATEGORY_DOMAINS, so policy

data is never injected. Template falsely says "We are developing a formal Supplier Code of Conduct"

Fix: Add buyer_requirements to POLICY_CATEGORY_DOMAINS (or create a specific mapping for supplier-related policies)

Q#: 55

Question (abbreviated): % suppliers ESG-assessed

Domain: buyer_requirements

Topics: supply_chain_monitoring

Template (line): L1333 supplier % KPI

Data?: Y

Issue: --

Fix:

Q#: 56

Question (abbreviated): Supply chain ESG monitoring

Domain: buyer_requirements

Topics: supply_chain_monitoring

Template (line): L1277 monitoring MEASURE

Data?: N

Issue: -- Generic but appropriate

Fix:

Q#: 57

Question (abbreviated): Responsible sourcing + conflict minerals (CMRT)

Domain: materials

Topics: conflict_minerals

Template (line): L1292 conflict minerals

Data?: N

Issue: MED: Only answers conflict minerals half; ignores "responsible sourcing policy for raw materials" part

Fix: Either split into two answers, or expand template to cover responsible sourcing before conflict minerals

---

Q#: 59

Question (abbreviated): Scope 3 transport + fleet + travel + commuting

Domain: emissions/transport

Topics: scope_3

Template (line): L148 scope 3

Data?: N

Issue: MED: Multi-part question — Scope 3 status answered but fleet composition, business travel km, and commuting

data

not addressed

Fix: Template should pull fleet/travel/commute data when transport is secondary domain; or question should be split

---

Q#: 58

Question (abbreviated): Supplier non-compliance handling

Domain: buyer_requirements

Topics: supply_chain_monitoring

Template (line): L1354 non-compliance MEASURE

Data?: N

Issue: --

Fix:

Q#: 60

Question (abbreviated): Transport emission reduction measures

Domain: transport

Topics: transport, logistics

Template (line): L1194 transport/logistics

Data?: N

Issue: LOW: Mentions carrier selection and route optimization but not fleet electrification, modal shift, or remote

work

Fix: Could add industry-specific transport reduction measures

---

Summary of Findings

By Severity

HIGH (3 questions) — answers wrong question or critical data gap:

Q#	Problem
37	Career development / succession planning question → answered with training hours. Completely wrong topic.
39	Human rights due diligence (MEASURE) → no template exists (HR template is POLICY-only). Falls to data dump.
52	External verification/assurance → no keyword rule, no template. Data fields (externalAssurance, assuranceStandard) exist but are orphaned.

| 54 | Supplier Code of Conduct → supplier_code policy IS active but buyer_requirements not in POLICY_CATEGORY_DOMAINS. Template falsely says "developing." |

MEDIUM (8 questions) — half-answered or data unused:

Q#	Problem
3	Revenue-only answer, ownership ignored
12	energySavingsKwh (38,700) exists but template doesn't use it
30	Raw recordable incident count (4) not surfaced, only TRIR rate
33	Confirms ISO 45001 but doesn't describe safety management approach as asked
36	Training types → same template as Q35, only 3 generic categories
38	Human rights answer cites H&S policy as human rights policy (misleading)

44	Anti-corruption training/monitoring/enforcement not addressed (same generic ethics template as Q43)	
50	Cert list has duplicates; validity dates available but not shown	
57	Only answers conflict minerals, skips responsible sourcing	
59	Multi-part: Scope 3 status only, fleet/travel/commute not addressed	

LOW (7 questions) — minor gaps or style issues:

Q8 (gas units), Q9 (methodology unnamed), Q13 (no roadmap detail), Q20 (no packaging data model), Q28 (DEI initiatives generic), Q43 (enforcement not covered), Q49 (risk mgmt generic), Q51 (latent bug: doesn't check flag), Q60 (transport measures generic)

Structural Issues

1. POLICY_CATEGORY_DOMAINS gap: buyer_requirements is not mapped, so supplier-related policies (supplier_code, supplier assessment) are invisible to templates in that domain. This is the root cause of Q54's false "developing" answer.

2. Training domain too coarse: Only one template covers all training questions. Q35 (hours KPI), Q36 (types), and Q37

(career development) all hit the same template. Need at least a MEASURE template for training types and a separate

template for development/succession.

3. Human rights MEASURE gap: The HR template only fires for POLICY questions. MEASURE-typed HR due diligence questions

(Q39) have no template to fall to.

4. External verification orphan: externalAssurance and assuranceStandard are collected in the profile, surfaced in

dataModel.ts, but no keyword rule routes to them and no template consumes them.

5. energySavingsKwh unused: Data is collected, bridged to the engine, but no template reads it. Q12 (energy efficiency

measures) is the natural consumer.

6. recordableIncidents not bridged: The dataBridge computes trirRate from incidents ÷ hours but doesn't pass the raw

incident count to CompanyData. Q30 asks for it explicitly.